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Education

- 2018 **PhD, Neuroscience**
Case Western Reserve University
Cleveland, OH
- 2010 **BSE, Biomedical Engineering**
Michigan Technological University
Houghton, MI

Employment

- 2024 - present **Tenure-Track Assistant Professor**
Department of Neuroscience
Joint appointment in the Department of Neurology
University of Wisconsin–Madison
Madison, WI
- 2017-2024 **Postdoctoral Scholar**
Laboratory of Prof. Ole Kiehn
Department of Neuroscience, University of Copenhagen
Copenhagen, Denmark
- 2010 - 2017 **Graduate Student**
Laboratories of Drs. Jerry Silver & Lynn T. Landmesser
Department of Neuroscience, Case Western Reserve University
Cleveland, OH

Scholarly Works

Highlighted Publications

1. **Cregg JM**[†], Sidhu SK, Leiras R, Kiehn O[†]. (2024) Basal ganglia-spinal cord pathway that commands locomotor gait asymmetries in mice. *Nature Neuroscience* 27:716-727. [†]Co-corresponding authors. ([pdf](#))
2. Leiras R*, **Cregg JM***, Kiehn O. (2022) Brainstem circuits for locomotion. *Annual Review of Neuroscience* 45:63-85. *Co-first authors. ([pdf](#))
3. **Cregg JM**, Leiras R, Montalant A, Wanken P, Wickersham IR, Kiehn O. (2020) Brainstem neurons that command mammalian locomotor asymmetries. *Nature Neuroscience* 23:730-740. ([pdf](#)) [Cover Article]
4. **Cregg JM**, Chu KA, Dick TE, Landmesser LT[†], Silver J[†]. (2017) Phasic inhibition as a mechanism for generation of rapid respiratory rhythms. *PNAS* 114:12815-12820. [†]Co-corresponding authors. ([pdf](#))
5. **Cregg JM**, Chu KA, Hager LE, Maggard RS, Stoltz DR, Edmond M, Alilain WJ, Philippidou P, Landmesser LT, Silver J. (2017) A latent propriospinal network can restore diaphragm function after high cervical spinal cord injury. *Cell Reports* 21:654-665. ([pdf](#))
6. Lang BT, **Cregg JM**, DePaul MA, Tran AP, Xu K, Dyck SM, Madalena KM, Brown BP, Weng YL, Li S, Karimi-Abdolrezaee S, Busch SA, Shen Y, Silver J. (2015) Modulation of the proteoglycan receptor PTP α promotes recovery after spinal cord injury. *Nature* 518:404-408. ([pdf](#))

Additional Publications

7. **Cregg JM**[†], Mirdamadi JL, Fortunato C, Okorokova EV, Kuper C, Nayeem R, Byun AJ, Avraham C, Buonocore A, Winner TS, Mildren RL. (2023) Highlights from the 31st Annual Meeting of the Society for the Neural Control of Movement. *Journal of Neurophysiology* 129:220-234. [†]Corresponding author. ([pdf](#))

8. Vagnozzi AN, Garg K, Dewitz C, Moore MT, **Cregg JM**, Jeannotte L, Zampieri N, Landmesser LT, Philippidou P. (2020) Phrenic-specific transcriptional programs shape respiratory motor output. *eLife* 9:e52859. ([pdf](#))
9. Lager AM, Corradin O, **Cregg JM**, Eliott MS, Shick E, Clayton BL, Allan KC, Olsen HE, Madhavan M, Tesar PJ. (2018) Rapid functional genetics of the oligodendrocyte lineage using pluripotent stem cells. *Nature Communications* 9:3708. ([pdf](#))
10. Niemi JP, DeFrancesco-Lisowitz A, **Cregg JM**, Howarth M, Zigmond RE. (2015) Overexpression of the monocyte chemokine CCL2 in dorsal root ganglion neurons causes a conditioning-like increase in neurite outgrowth and does so via a STAT3 dependent mechanism. *Experimental Neurology* 275:25-37. ([pdf](#))
11. Gardner RT, Wang L, Lang BT, **Cregg JM**, Dunbar CL, Woodward WR, Silver J, Ripplinger CM, Habecker BA. (2015) Targeting protein tyrosine phosphatase sigma after myocardial infarction restores cardiac sympathetic innervation and prevents arrhythmias. *Nature Communications* 6:6235. ([pdf](#))
12. **Cregg JM**, DePaul MA, Filous AR, Lang BT, Tran A, Silver J. (2014) Functional regeneration beyond the glial scar. *Experimental Neurology* 253:197-207. ([pdf](#))
13. Hilton BJ, Lang BT, **Cregg JM**. (2012) Keratan sulfate proteoglycans in plasticity and recovery after spinal cord injury. *Journal of Neuroscience* 32:4331-4333. ([pdf](#))
14. Hurtado A*, **Cregg JM***, Wang HB, Wendell DF, Oudega M, Gilbert RJ, McDonald JW. (2011) Robust CNS regeneration after complete spinal cord transection using aligned poly-L-lactic acid microfibers. *Biomaterials* 32:6068-6079. *Co-first authors. ([pdf](#))
15. Wang HB, Mullins ME, **Cregg JM**, McCarthy CM, Gilbert RJ. (2010) Varying the diameter of aligned electrospun fibers alters neurite outgrowth and Schwann cell migration. *Acta Biomaterialia* 6:2970-2978. ([pdf](#))
16. **Cregg JM**, Wiseman SL, Pietrzak-Goetze NM, Smith MR, Jaroch DB, Clupper DL, Gilbert RJ. (2010) A rapid, quantitative method for assessing axonal extension on biomaterial platforms. *Tissue Engineering Part C: Methods* 16:167-172. ([pdf](#)) [Cover Article]
17. Wang HB, Mullins ME, **Cregg JM**, Hurtado A, Oudega M, Trombley MT, Gilbert RJ. (2009) Creation of highly aligned electrospun poly-L-lactic acid fiber for nerve regeneration applications. *Journal of Neural Engineering* 6:016001. ([pdf](#)) [Cover Article]

Preprints

1. Pende M^{*†}, **Cregg JM***, Saghaei S, Broadbent S, Avdibasic A, Roeles J, Papadopoulos SC, Seaman RP, Pende N, Solano Mateos M, Jamwal K, Wunch M, Pasierbek P, Moreno-Cencerrado A, Korchynska S, Hauer R, Anderson P, Supper P, Kastri ME, Reumann D, Moorhead M, Graber JH, Scholze P, Henschke JU, Budinger E, Knoblich JA, Klausberger T, Adameyko I, Harkany T, Kumar V, Joy MT, Kiehn O, Dodt HU[†], Voigt FF[†], Murawala P[†]. (2026) A uniform tissue-clearing framework and mesoSPIM-ultra enable cm-scale single-neuron tracing. *BioRxiv*. *Co-first authors. [†]Co-corresponding authors. ([pdf](#))

Patents

1. Hurtado A, Gilbert RJ, Wang HB, **Cregg JM**, Mullins ME, Oudega M. (2019) Three-dimensional scaffolds, methods for fabricating the same, and methods of treating a peripheral nerve or spinal cord injury. US Patent 10,413,391. ([pdf](#))
2. Silver J, Lang BT, **Cregg JM**, Weng YL, Li H, Wu W. (2019) Compositions and methods of treating root avulsion injury. US Patent 10,258,672. ([pdf](#))
3. Lang BT, **Cregg JM**, Weng YL, Silver J. (2018) Compositions and methods for inhibiting the activity of lar family phosphatases. US Patent 9,937,242. ([pdf](#)) [Licensed to NervGen Pharma]

Bibliometric Summary

Web of Science: >2200 citations, h-index 14 ([link](#))

Google Scholar: >3500 citations, h-index 17 ([link](#))

Funding

2026 - 2029	Edward Mallinckrodt, Jr. Foundation Grant Award	\$225,000 USD
2025 - 2028	Parkinson's Foundation - Stanley Fahn Junior Faculty Award	\$300,000 USD
2025 - 2027	Brain Research Foundation - Seed Grant	\$100,000 USD

Honors & Fellowships

- 2026 C.J. Herrick Award in Neuroanatomy, American Association for Anatomy
- 2025 Stanley Fahn Junior Faculty Award, Parkinson's Foundation
- 2021 Lundbeck Foundation Postdoctoral Fellowship
- 2018 EMBO Long-Term Fellowship
- 2018 Doctoral Excellence Award in Neurosciences, Case Western Reserve University
- 2010 NSF Graduate Research Fellowship
- 2008 Barry M. Goldwater Scholarship

Invited/Conference Talks

- 2026 Anatomy Connected Meeting, *Albuquerque, NM*
- 2026 Embodied Intelligence Conference, Plenary Lecture, *Virtual*
- 2026 Parkinson's Foundation Research Summit, *Atlanta, GA*
- 2026 EPFL, Ecole Polytechnique Fédérale de Lausanne, Institute of Bioengineering, *Lausanne, Switzerland*
- 2025 Temple University, Department of Neural Sciences, *Philadelphia, PA*
- 2025 Case Western Reserve University, Department of Neuroscience, *Cleveland, OH*
- 2025 University of Copenhagen, Faculty of Health and Medical Sciences, *Copenhagen, Denmark*
- 2025 Marquette University, Department of Biomedical Engineering, *Milwaukee, WI*
- 2024 BIG10 Neuroscience Seminar Series, *Virtual*
- 2024 MDI Biological Laboratory, *Bar Harbor, ME*
- 2024 European Molecular Biology Laboratory, DANEMO Symposium, *Copenhagen, Denmark*
- 2024 California Institute of Technology, New Horizons in Neuroscience Symposium, *Pasadena, CA*
- 2023 University of Wisconsin–Madison, Department of Neurology, Neurology Grand Rounds, *Madison, WI*
- 2023 Benzon Symposium: Bringing Circuit for Movement Together, *Copenhagen, Denmark*
- 2023 University of Minnesota, Department of Neuroscience, *Minneapolis, MN*
- 2023 Motor Control: Spinal Circuits and Beyond, *St Andrews, Scotland*
- 2023 XIV Meeting of the International Basal Ganglia Society, *Stockholm, Sweden*
- 2023 Karolinska Institutet, Department of Neuroscience, *Stockholm, Sweden*
- 2023 University of St Andrews, School of Psychology and Neuroscience, *St Andrews, Scotland*
- 2023 Stony Brook University, Department of Neurobiology and Behavior, *Stony Brook, NY*
- 2023 Yale University, Department of Neuroscience, *New Haven, CT*
- 2022 Case Western Reserve University, Department of Neuroscience, *Cleveland, OH*
- 2022 Annual Meeting of the Society for the Neural Control of Movement, *Dublin, Ireland*
- 2022 Basal Ganglia Gordon Research Seminar, *Ventura, CA*
- 2021 Danish Society for Neuroscience, Brain States Meeting, *Copenhagen, Denmark*
- 2020 Sainsbury Wellcome Center, Emerging Neuroscientists Seminar Series, *London, United Kingdom*
- 2020 International Online Spinal Cord Injury Research Seminars, *Virtual*
- 2019 Okinawa Institute of Science & Technology, Neuronal Circuits in Motor Behavior, *Okinawa, Japan*
- 2016 National Neurotrauma Society Annual Meeting, *Lexington, KY*
- 2015 Case Western Reserve University, Dept. of Pulmonary, Critical Care, and Sleep Medicine, *Cleveland, OH*
- 2010 Society for Biomaterials Annual Meeting, *Seattle, WA*
- 2008 Biomedical Engineering Society Annual Meeting, *St. Louis, MO*

Mentoring

PhD Students

- Daniel J. Bader, 2025 - present, Neuroscience Training Program, UW–Madison
- Anusha Shukla, 2024 - present, Biophysics Training Program, UW–Madison
- Songyuan Tan, 2024 - present, Neuroscience Training Program, UW–Madison

MS Students

- Simrandeep K. Sidhu, MS Thesis in Neuroscience, 2020-2023, University of Copenhagen
- Paulina Wanken, 2018 - 2020, MS Thesis in Human Biology, University of Copenhagen

Undergraduate Students

- Finn E. Gustafson, 2026 - present, UW–Madison, Neurobiology
- Audrey A. Zeller, 2026 - present, UW–Madison, Biomedical Engineering

- Yuhua Yang, 2026, UW–Madison, Biochemistry/Psychology
- Karina L. Fischer, 2026 - present, UW–Madison, Neurobiology/Biochemistry
- Utkarsha Marasini, 2025 - present, UW–Madison, Mechanical Engineering
- Areeb T. Bajwa, 2024 - 2025, UW–Madison, Biochemistry
- Amelia E. Lehmann, BS, 2024 - 2025, UW–Madison, Neurobiology/Biochemistry
- Aleksandar Zafirovski, 2024 - 2026, UW–Madison, Neurobiology
- Kevin A. Chu, BS Thesis in Biology, 2015 - 2017, Case Western Reserve University

Teaching

University of Wisconsin–Madison

- 2026 - 2027 Course Director, NEURO747: Sensory and Motor Systems
- 2025 - 2026 Course Director, PT500: Functional Neuroanatomy
- 2024 - 2026 Guest Lecturer, Med SC-M 775: Mind & Motion
- 2024 - 2025 Guest Lecturer, Zoology 500: Undergraduate Neurobiology Seminar

University of Copenhagen

- 2023 - 2024 Lecturer, Open Neurophysiology: Analysis Tools & Datasets (PhD); Course Co-organizer, 2024
- 2022 Guest Lecturer, Animal Models of Disease and Behavior (PhD)
- 2021 Guest Lecturer, Workshop on Animal Models
- 2018 - 2022 Guest Lecturer, Neuronal Signaling/Neuroscience (MS)

Neuroscience Academy Denmark

- 2024 Guest Lecturer, Translational Neuroscience (PhD)

Case Western Reserve University

- 2017 Section Instructor, PHOL519: Cardiorespiratory Physiology
- 2017 Section Instructor, PHOL466: Cell Signaling

Service

Committees

- Research Laser Safety Committee, UW–Madison (2026 - present)
- Selection Committee – Molecular Biophysics T32 Training Program, UW–Madison (2025)
- Seminar Committee – Neuroscience Seminar Series, UW–Madison (2025 - present)
- Admissions Committee – Biophysics Training Program, UW–Madison (2024 - present)
- Admissions Committee – Neuroscience Training Program, UW–Madison (2024 - 2026)

Peer Review

Journals

Ad hoc reviewer for *Nature*, *Science*, *Experimental Neurology*, *Scientific Reports*

Grants

- Review Committee – Parkinson's Foundation, Stanley Fahn Junior Faculty Award (2026 - 27)
- Review Committee – Parkinson's Foundation, Launch Award Program (2026)
- Review Committee – Parkinson's Foundation, Postdoctoral Fellowships (2025)
- Review Committee – UW–Madison Brain Research Foundation Seed Grant Program internal competition (Office of the Vice Chancellor for Research and Graduate Education, 2025)
- Ad hoc reviewer for *Swiss National Science Foundation*, *UW–Madison Institute for Clinical and Translational Research*